

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [2]: list_of_dicts=[{'name':'Ginger','breed':'Dachshund','height_cm':22,'weight_kg':10,'
                        {'name':'Scout','breed':'Dalmatian','height_cm':59,'weight_kg':25,'
                        ]
```

```
In [3]: new_dogs=pd.DataFrame(list_of_dicts)
new_dogs
```

```
Out[3]:
```

	name	breed	height_cm	weight_kg	date_of_birth
0	Ginger	Dachshund	22	10	2019-03-14
1	Scout	Dalmatian	59	25	2019-05-09

```
In [4]: dicts_of_lists={
        'name':['Ginger','Scout'],
        'breed':['Dachshund','Dalmaintain'],
        'height_cm':[22,59],
        'weight_kg':[10,25],
        'date_of_birth':['2019-03-14','2019-05-09']
    }
```

```
In [5]: new_dogs=pd.DataFrame(dicts_of_lists)
new_dogs
```

```
Out[5]:
```

	name	breed	height_cm	weight_kg	date_of_birth
0	Ginger	Dachshund	22	10	2019-03-14
1	Scout	Dalmaintain	59	25	2019-05-09

```
In [6]: avocado=pd.read_csv(r"C:\Users\Lenovo\Documents\Notes\APRIL\4th - REGRESSION PROJEC
avocado.head()
```

Out[6]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

In [7]:

```
avocado=avocado.reset_index(drop=True)
avocado.head()
```

Out[7]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

In [8]:

```
avocado.to_csv("test_write.csv")
```

In [9]:

```
avocado=pd.read_csv(r"C:\Users\Lenovo\Documents\Notes\APRIL\4th - REGRESSION PROJEC
avocado.head()
```

Out[9]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
0	0	2015-12-27	1.33	64236.62	1036.74	54454.85	48.16	8696.87	8603.62
1	1	2015-12-20	1.35	54876.98	674.28	44638.81	58.33	9505.56	9408.07
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35	8042.21
3	3	2015-12-06	1.08	78992.15	1132.00	71976.41	72.58	5811.16	5677.40
4	4	2015-11-29	1.28	51039.60	941.48	43838.39	75.78	6183.95	5986.26

In [10]:

avocado.tail()

Out[10]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small B
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13060
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9350
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10910
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11980

In [11]:

avocado.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18249 entries, 0 to 18248
Data columns (total 14 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            18249 non-null  int64
1   Date                  18249 non-null  object
2   AveragePrice          18249 non-null  float64
3   Total Volume         18249 non-null  float64
4   4046                  18249 non-null  float64
5   4225                  18249 non-null  float64
6   4770                  18249 non-null  float64
7   Total Bags           18249 non-null  float64
8   Small Bags           18249 non-null  float64
9   Large Bags           18249 non-null  float64
10  XLarge Bags          18249 non-null  float64
11  type                 18249 non-null  object
12  year                 18249 non-null  int64
13  region               18249 non-null  object
dtypes: float64(9), int64(2), object(3)
memory usage: 1.9+ MB
```

```
In [12]: print(avocado.shape)

(18249, 14)
```

```
In [13]: avocado.describe()
```

Out[13]:

	Unnamed: 0	AveragePrice	Total Volume	4046	4225	477
count	18249.000000	18249.000000	1.824900e+04	1.824900e+04	1.824900e+04	1.824900e+0
mean	24.232232	1.405978	8.506440e+05	2.930084e+05	2.951546e+05	2.283974e+0
std	15.481045	0.402677	3.453545e+06	1.264989e+06	1.204120e+06	1.074641e+0
min	0.000000	0.440000	8.456000e+01	0.000000e+00	0.000000e+00	0.000000e+0
25%	10.000000	1.100000	1.083858e+04	8.540700e+02	3.008780e+03	0.000000e+0
50%	24.000000	1.370000	1.073768e+05	8.645300e+03	2.906102e+04	1.849900e+0
75%	38.000000	1.660000	4.329623e+05	1.110202e+05	1.502069e+05	6.243420e+0
max	52.000000	3.250000	6.250565e+07	2.274362e+07	2.047057e+07	2.546439e+0

```
In [14]: print(avocado.columns)

Index(['Unnamed: 0', 'Date', 'AveragePrice', 'Total Volume', '4046', '4225',
      '4770', 'Total Bags', 'Small Bags', 'Large Bags', 'XLarge Bags', 'type',
      'year', 'region'],
      dtype='object')
```

```
In [15]: even=pd.Series([2,4,6,8,10])
         odd=pd.Series([1,3,5,7,9])
```

```
res=even.append(odd)
res
```

```
-----
AttributeError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_11092\2012875665.py in ?()
      1 even=pd.Series([2,4,6,8,10])
      2 odd=pd.Series([1,3,5,7,9])
      3
----> 4 res=even.append(odd)
      5 res

~\anaconda3\Lib\site-packages\pandas\core\generic.py in ?(self, name)
    6295         and name not in self._accessors
    6296         and self._info_axis._can_hold_identifiers_and_holds_name(name)
    6297     ):
    6298         return self[name]
-> 6299     return object.__getattr__(self, name)

AttributeError: 'Series' object has no attribute 'append'
```

```
In [49]: even = pd.Series([2,4,6,8,10])
        odd = pd.Series([1,3,5,7,9])

        res = pd.concat([even,odd])
        res
```

```
Out[49]: 0      2
         1      4
         2      6
         3      8
         4     10
         0      1
         1      3
         2      5
         3      7
         4      9
        dtype: int64
```

```
In [51]: res.reset_index(drop=True)
```

```
Out[51]: 0      2
         1      4
         2      6
         3      8
         4     10
         5      1
         6      3
         7      5
         8      7
         9      9
        dtype: int64
```

```
In [53]: avocado.sort_values(["AveragePrice", "year"], ascending=[True, False])
```

Out[53]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	
15261	43	2017-03-05	0.44	64057.04	223.84	4748.88	0.00	5908
7412	47	2017-02-05	0.46	2200550.27	1200632.86	531226.65	18324.93	45036
15473	43	2017-03-05	0.48	50890.73	717.57	4138.84	0.00	4605
15262	44	2017-02-26	0.49	44024.03	252.79	4472.68	0.00	3929
1716	0	2015-12-27	0.49	1137707.43	738314.80	286858.37	11642.46	10089
...	...	...	...	...	...	...	...	...
16720	18	2017-08-27	3.04	12656.32	419.06	4851.90	145.09	724
16055	42	2017-03-12	3.05	2068.26	1043.83	77.36	0.00	94
14124	7	2016-11-06	3.12	19043.80	5898.49	10039.34	0.00	310
17428	37	2017-04-16	3.17	3018.56	1255.55	82.31	0.00	168
14125	8	2016-10-30	3.25	16700.94	2325.93	11142.85	0.00	325

18249 rows × 14 columns



In [55]:

avocado['AveragePrice']

Out[55]:

01234

1.331.350.931.081.28

...

1824418245182461824718248

1.631.711.871.931.62

Name: AveragePrice, Length: 18249, dtype: float64

In [57]:

avocado[['AveragePrice']]<1

Out[57]:

	AveragePrice
0	False
1	False
2	True
3	False
4	False
...	...
18244	False
18245	False
18246	False
18247	False
18248	False

0	False
1	False
2	True
3	False
4	False
...	...
18244	False
18245	False
18246	False
18247	False
18248	False

18249 rows × 1 columns

```
In [59]: avocado[avocado['AveragePrice']<1]
```

Out[59]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags
2	2	2015-12-13	0.93	118220.22	794.70	109149.67	130.50	8145.35
6	6	2015-11-15	0.99	83453.76	1368.92	73672.72	93.26	8318.86
7	7	2015-11-08	0.98	109428.33	703.75	101815.36	80.00	6829.22
13	13	2015-09-27	0.99	106803.39	1204.88	99409.21	154.84	6034.46
43	43	2015-03-01	0.99	55595.74	629.46	45633.34	181.49	9151.45
...	...	...	...	...	...	...	...	...
17169	43	2017-03-05	0.99	155011.12	35367.23	5175.81	5.91	114462.17
17170	44	2017-02-26	0.99	171145.00	34520.03	6936.39	0.00	129688.58
17536	39	2017-04-02	0.98	402676.23	34093.33	58330.53	207.85	310044.52
17537	40	2017-03-26	0.90	456645.91	36169.35	51398.72	139.55	368938.29
17540	43	2017-03-05	0.99	367519.17	61166.48	55123.99	126.80	251101.90

2796 rows × 14 columns



In [61]: avocado[avocado['type']=='organic']

Out[61]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Sr B
9126	0	2015-12-27	1.83	989.55	8.16	88.59	0.00	892.80	890
9127	1	2015-12-20	1.89	1163.03	30.24	172.14	0.00	960.65	960
9128	2	2015-12-13	1.85	995.96	10.44	178.70	0.00	806.82	800
9129	3	2015-12-06	1.84	1158.42	90.29	104.18	0.00	963.95	940
9130	4	2015-11-29	1.94	831.69	0.00	94.73	0.00	736.96	730
...	...	...	...	...	...	...	...	...	...
18244	7	2018-02-04	1.63	17074.83	2046.96	1529.20	0.00	13498.67	13060
18245	8	2018-01-28	1.71	13888.04	1191.70	3431.50	0.00	9264.84	8940
18246	9	2018-01-21	1.87	13766.76	1191.92	2452.79	727.94	9394.11	9350
18247	10	2018-01-14	1.93	16205.22	1527.63	2981.04	727.01	10969.54	10910
18248	11	2018-01-07	1.62	17489.58	2894.77	2356.13	224.53	12014.15	11980

9123 rows × 14 columns



In [63]: avocado[avocado['Date']<='2015-02-04']

Out[63]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	S
47	47	2015-02-01	0.99	70873.60	1353.90	60017.20	179.32	9323.18	91
48	48	2015-01-25	1.06	45147.50	941.38	33196.16	164.14	10845.82	101
49	49	2015-01-18	1.17	44511.28	914.14	31540.32	135.77	11921.05	116
50	50	2015-01-11	1.24	41195.08	1002.85	31640.34	127.12	8424.77	80
51	51	2015-01-04	1.22	40873.28	2819.50	28287.42	49.90	9716.46	91
...	...	...	...	...	...	...	...	...	...
11928	46	2015-02-01	1.77	7210.19	1634.42	3012.44	0.00	2563.33	25
11929	47	2015-01-25	1.63	7324.06	1934.46	3032.72	0.00	2356.88	23
11930	48	2015-01-18	1.71	5508.20	1793.64	2078.72	0.00	1635.84	16
11931	49	2015-01-11	1.69	6861.73	1822.28	2377.54	0.00	2661.91	26
11932	50	2015-01-04	1.64	6182.81	1561.30	2958.17	0.00	1663.34	16

540 rows × 14 columns



In [65]: avocado[(avocado['Date']<='2015-02-04')& (avocado['type']=='organic')]

Out[65]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags
9173	47	2015-02-01	1.83	1228.51	33.12	99.36	0.0	1096.03	1096.03
9174	48	2015-01-25	1.89	1115.89	14.87	148.72	0.0	952.30	952.30
9175	49	2015-01-18	1.93	1118.47	8.02	178.78	0.0	931.67	931.67
9176	50	2015-01-11	1.77	1182.56	39.00	305.12	0.0	838.44	838.44
9177	51	2015-01-04	1.79	1373.95	57.42	153.88	0.0	1162.65	1162.65
...	...	...	...	...	...	...	...	...	...
11928	46	2015-02-01	1.77	7210.19	1634.42	3012.44	0.0	2563.33	2563.33
11929	47	2015-01-25	1.63	7324.06	1934.46	3032.72	0.0	2356.88	2320.00
11930	48	2015-01-18	1.71	5508.20	1793.64	2078.72	0.0	1635.84	1620.00
11931	49	2015-01-11	1.69	6861.73	1822.28	2377.54	0.0	2661.91	2656.66
11932	50	2015-01-04	1.64	6182.81	1561.30	2958.17	0.0	1663.34	1663.34

270 rows × 14 columns



```
In [67]: regionFilter=avocado['region'].isin(['Boston','SanDiego'])
avocado[regionFilter]
```

Out[67]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags
208	0	2015-12-27	1.13	450816.39	3886.27	346964.70	13952.56	86012.86
209	1	2015-12-20	1.07	489802.88	4912.37	390100.99	5887.72	88901.80
210	2	2015-12-13	1.01	549945.76	4641.02	455362.38	219.40	89722.96
211	3	2015-12-06	1.02	488679.31	5126.32	407520.22	142.99	75889.78
212	4	2015-11-29	1.19	350559.81	3609.25	272719.08	105.86	74125.62
...	...	...	...	...	...	...	...	...
17656	7	2018-02-04	1.83	28934.01	9.14	2504.27	3.66	26416.94
17657	8	2018-01-28	1.72	30846.35	14.65	1628.45	0.00	29203.25
17658	9	2018-01-21	1.82	26704.98	60.87	2647.48	0.00	23996.63
17659	10	2018-01-14	1.73	29468.36	12.77	2380.23	0.00	27075.36
17660	11	2018-01-07	1.91	30096.00	14.46	2704.48	0.00	27377.06

338 rows × 14 columns



```
In [69]: regionFilter=avocado['region'].isin(['Boston','SanDiego'])
yearFilter=avocado['year'].isin(['2016','2017'])
avocado[regionFilter&yearFilter]
```

Out[69]:

	Unnamed: 0	Date	AveragePrice	Total Volume	4046	4225	4770	Total Bags	Small Bags	Large Bags	XLarge Bags

```
In [ ]: avocado.isna()
```

```
In [ ]: avocado.isna().any()
```

```
In [ ]: avocado.isna().sum()
```

```
In [ ]: avocado.dropna()
```

```
In [ ]: meanVal=avocado['AveragePrice'].mean()
        avocado.fillna(meanVal)

In [ ]: avocado['AveragePricePer100']=avocado['AveragePrice']*100
        avocado

In [ ]: avocado.drop(['AveragePricePer100'],axis=1)

In [ ]: avocado["AveragePrice"].mean()

In [ ]: avocado['Date'].max()

In [ ]: def pct30(column):
        return column.quantile(0.3)
        def pct50(column):
        return column.quantile(0.5)

        avocado[['AveragePrice','Total Bags']].agg([pct30,pct50])

In [ ]: temp=avocado.drop_duplicates(subset=['year'])
        temp

In [ ]: avocado['year'].value_counts(sort=True,ascending=False)

In [ ]: avocado.pivot_table(index=['year','type'],aggfunc=[min,max,np.mean,np.median],value

In [ ]: regionIndex=avocado.set_index(['region'])
        regionIndex

In [ ]: avocado[avocado['region'].isin(['Albany','WestTexNewMexico'])]

In [ ]: regionIndex.loc[['Albany','WestTexNewMexico']]

In [ ]: avocado["AveragePrice"].hist(bins=20)

In [ ]: regionFilter=avocado.groupby('region')['AveragePrice'].mean().head(10)
        regionFilter

In [ ]: regionFilter.plot(kind='bar',rot=45,title='Average price in 10 grams')

In [ ]: avocado.plot(x='AveragePrice',y='Total Volume',kind='scatter')

In [ ]: avocado['AveragePrice'].sub(avocado['AveragePrice'])

In [ ]:

In [ ]:

In [ ]:
```

In []: